PLATFORMDAY · OPEN SOURCE AI INFRASTRUCTURE

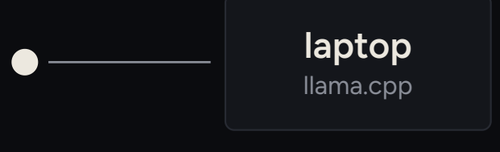
vLLM Explained

What an open source inference server actually does, between your app and the GPU.

Raul Leite rauldsl.com Views are my own

Running a model is easy.
Serving it is hard.

ONE USER



fast enough · cheap · done

HUNDREDS OF USERS



~10×

2023 estimate: cost of one LLM request vs. one keyword search

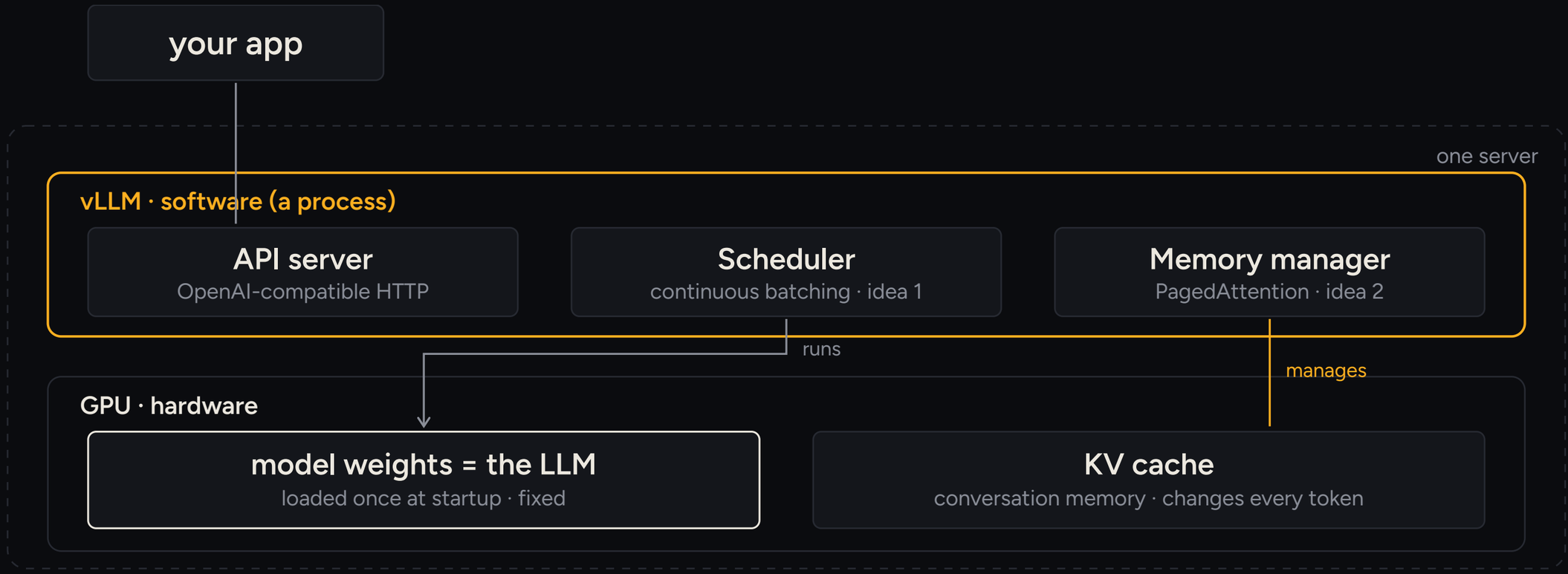
Analyst estimate cited in Kwon et al., SOSP '23 · summarized in R. Leite, InfoWorld, Sep 2025

Something has to sit between your app and the GPU.

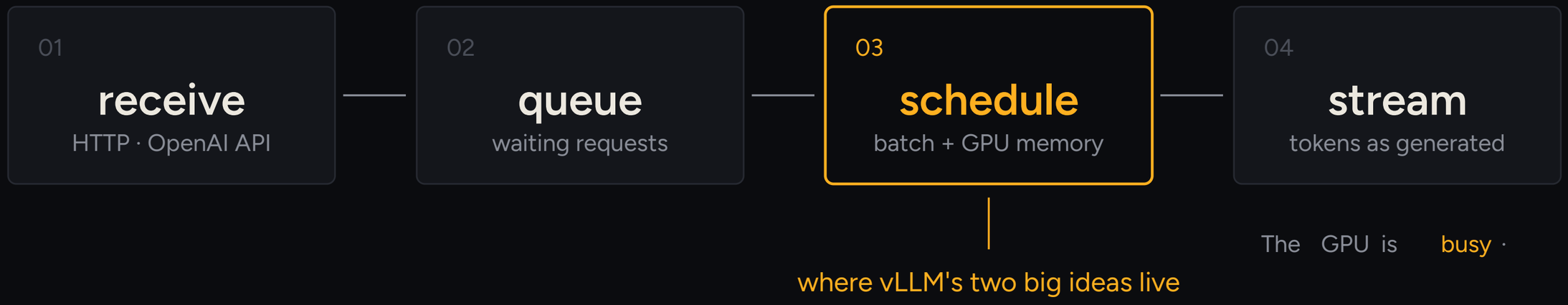


the piece platform teams now own

The LLM lives on the GPU. vLLM runs it.



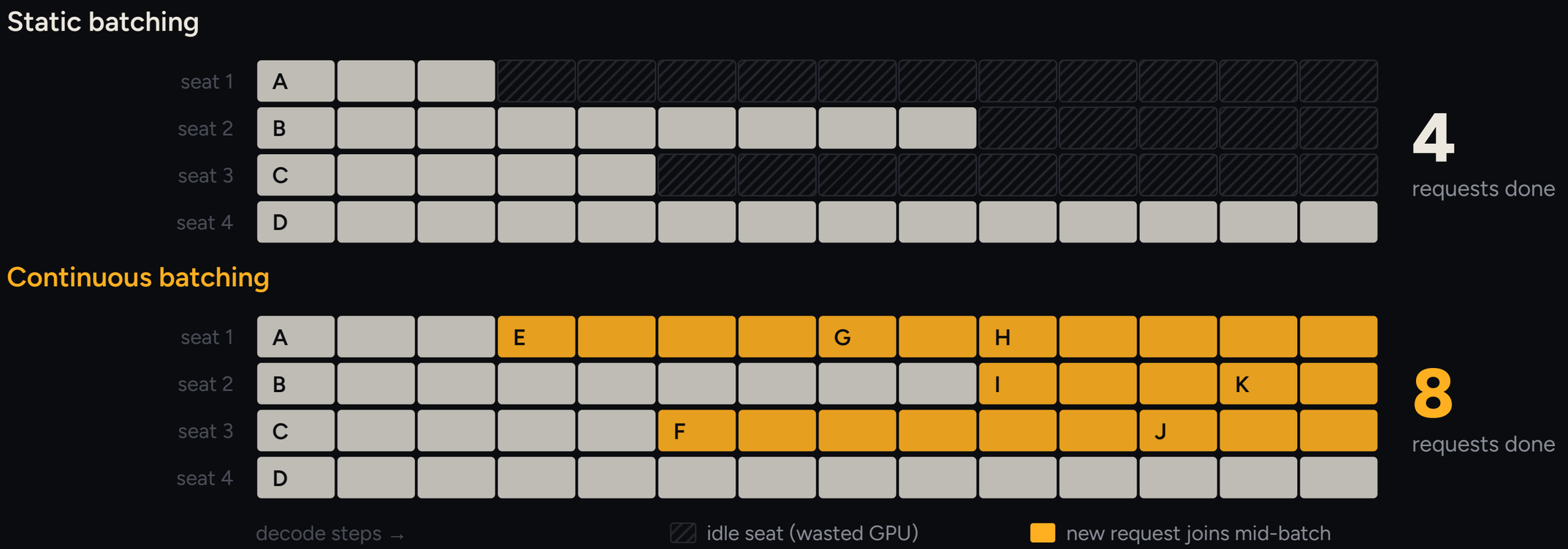
An inference server does four things.



IDEA 1 · CONTINUOUS BATCHING

Don't wait for the slowest request.

vLLM schedules **one token step at a time** and refills a seat the moment it frees up.



illustration

IDEA 2 · PAGEDATTENTION

Manage GPU memory like an OS manages pages.

Contiguous reservations

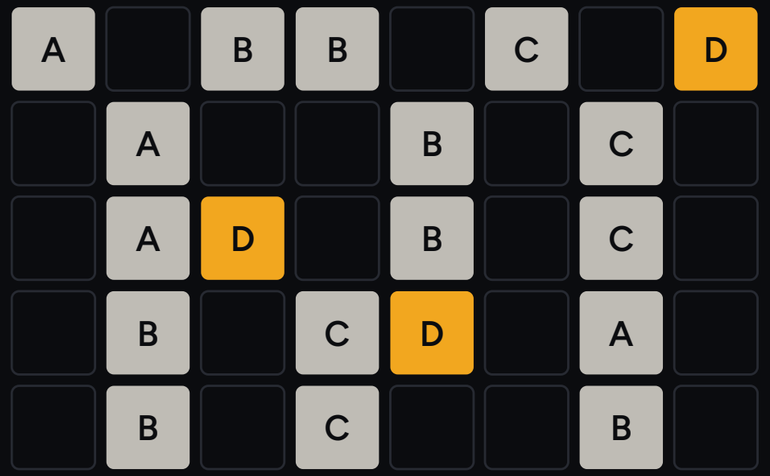
each request reserves its max length



D waits: 10 cells free, never 10 in a row

Paged KV blocks

fixed-size blocks, allocated on demand



D fits: blocks don't need to be adjacent

KV block is like a **page**

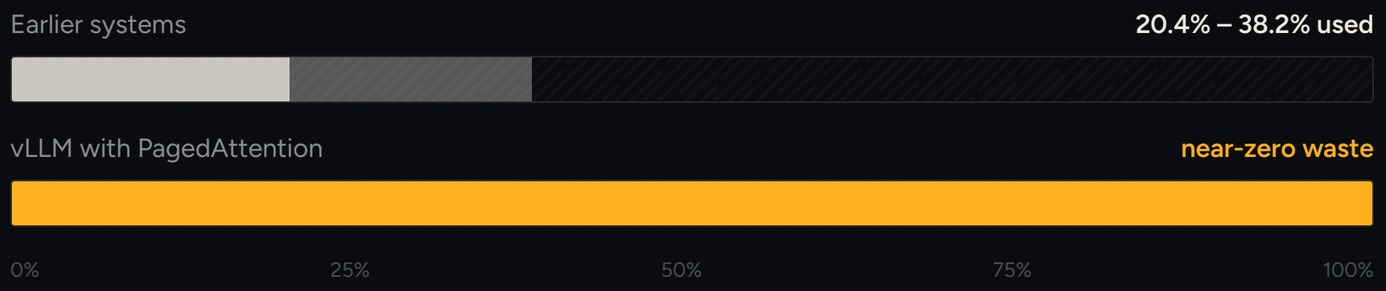
token is like a **byte**

request is like a **process**

05 · THE PAYOFF

20–38%

of KV cache memory actually held tokens in earlier serving systems



2–4×

throughput vs. FasterTransformer and Orca, without increasing latency

Source: Kwon et al., “Efficient Memory Management for LLM Serving with PagedAttention,” SOSP ’23, as summarized in R. Leite, InfoWorld, Sep 2025

Your app changes one line.

SERVER

```
vllm serve Qwen/Qwen2.5-7B-Instruct   # OpenAI-compatible API on :8000
```

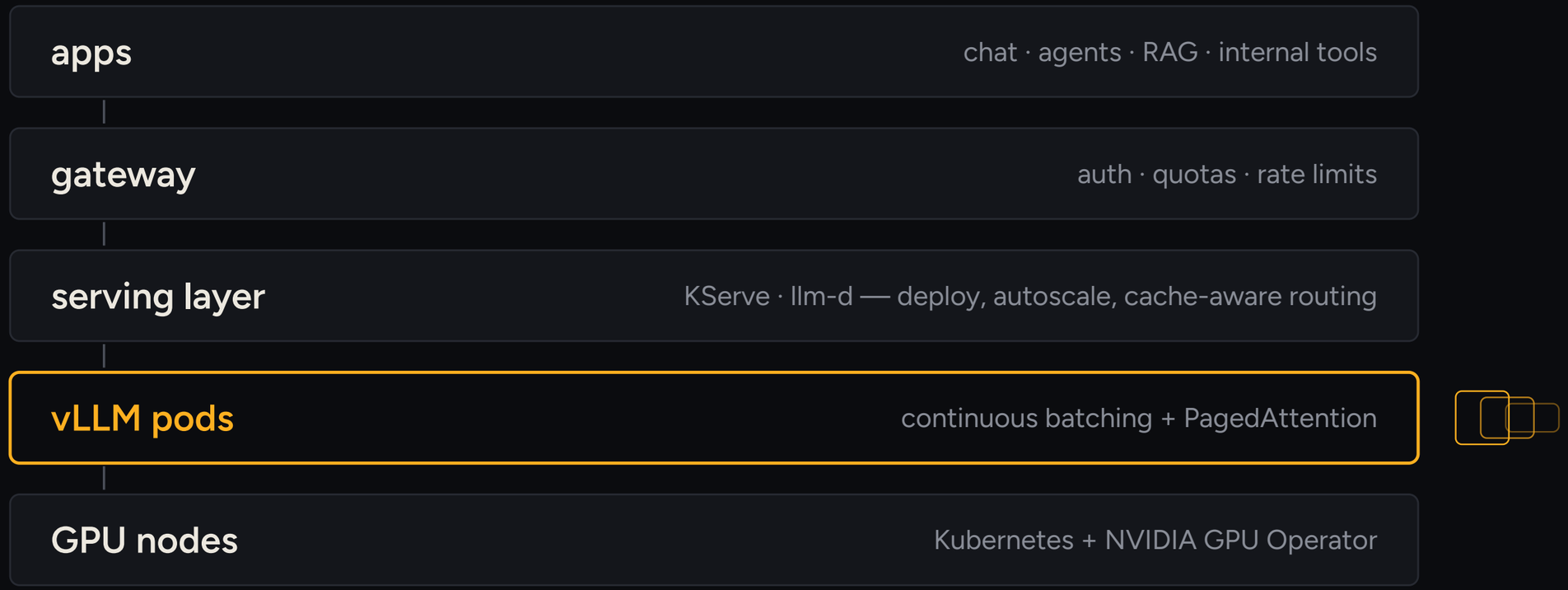
YOUR APP, SAME OPENAI CLIENT

```
from openai import OpenAI

client = OpenAI(
    base_url="http://vllm:8000/v1",
    api_key="unused",
)
client.chat.completions.create(
    model="Qwen/Qwen2.5-7B-Instruct",
    messages=[{"role": "user", "content": "Hi"}],
)
```

07 · THE PLATFORM

On Kubernetes, it's just another workload.



08 · THE CHOICE

Pick the tool for the scale.

ONE USER

llama.cpp · Ollama

Laptop, dev loop, offline. Runs on CPU or an Apple GPU.

Laptop

MANY USERS

vLLM

One GPU server, many concurrent requests, OpenAI-compatible API.

A FLEET

vLLM on Kubernetes

Many replicas and models, scaled and routed with KServe or llm-d.

Platform

What you get.

2–4×

more users on the same GPU

Less GPU

lower cost for the same demand

Your data

open models running inside your walls

1 line

same OpenAI API · no rewrite, no lock-in

2–4× throughput: Kwon et al., SOSP '23

10 · TAKEAWAY

Your platform already knows how to do this.

Same GPU. More users.
Your data. Your Kubernetes.

Raul Leite
raulds l.com
linkedin.com/in/raul-leite
InfoWorld · PagedAttention & continuous batching